

Groundwork for Neural Network-Based Specific Emitter Identification Authentication for IoT

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Abstract—Trust is a prominent concern with the continued expansion of the Internet of Things (IoT). As new devices enter the market, device security must be a design pillar. In order to trust these devices, they must be identifiable and authenticated before they begin transmitting possible sensitive information, and given the vast number of IoT devices in the future, it may prove difficult to properly trust and authenticate these authorized devices on networks with current methods. Machine learning neural networks have the ability to uniquely identify transmitters based on their physical waveform characteristics which could be used to identify and authenticate IoT nodes in large networks with little impact to latency, providing an extra layer of security and trust. This paper presents the groundwork for performing neural network-based specific emitter identification (SEI) on resource constrained IoT devices using only raw in-phase and quadrature (IQ) streams, with protocols to secure IoT networks. As proof of concept, an existing neural network-based SEI algorithm is executed on both a resource-rich and a more resource-constrained device with low latency, demonstrating the feasibility of using such algorithms on IoT devices now and in the future.

Index Terms—Internet of Things, Security, Authentication, Neural Networks, Specific Emitter Identification, RF Fingerprinting

I. INTRODUCTION

The Internet of Things (IoT) is becoming more commonplace in our everyday lives and experts expect the number of these devices to grow past 50 billion by the year 2020 [1]. The IoT aims to improve many aspects of our lives through efficiency. For example, smart homes, smart vehicles, and smart city infrastructures are being designed to reduce energy costs by incorporating numerous sensors and devices that can automatically change their behavior to better react to our behavioral patterns [2]. Further, the healthcare industry is turning to IoT-based solutions for in-home and personal care [3], and industrial IoT (IIoT) networks are being utilized to improve product manufacturing, production, and storage. These IoT networks are typically well-defined and hierarchical in nature [4], and are configured for specific purposes, such as a factory floor layout for manufacturing [5]. Such general network topologies help reduce many unknown factors of wireless sensor networks (WSNs) by limiting the size, mobility, and communication routing of IoT devices. However, security requirements for these networks is not diminished. Conversely,

as the IoT continues to grow, security will become a bigger concern as these devices will have access to and provide personal and/or private information about people, vehicles, and critical structures and systems.

The size and flexibility of the IoT brings other unique security concerns even when traditional security schemes are properly implemented. For example, the scalability of IoT increases the possibility for remote device or data observation, allowing an attacker to collect vast amounts of information to help in reverse engineering of security protocols [6]. The use of authentication techniques in such a network ensures that only trusted devices are able to communicate on a network. However, most authentication protocols currently used only perform non-physical data inspection (i.e. data content, not physical signal), similar to Data Encryption Standard (DES) [7], Wi-Fi Protected Access (WPA) [8], WPA2 [9], or elliptic curve cryptography (ECC) [10], harboring latent defects. More specifically, they fail to authenticate the actual device which can lead to possible device compromise or impersonation, as all of the involved parameters can be observed, guessed, or replicated [11]. As the number of small, poorly authenticated, wireless IoT devices continues to grow, common network attacks such as eavesdropping, man-in-the-middle (MITM), denial of service (DoS), node impersonation, and battery draining [12], [13], [14], will only become more prevalent, if security continues to be neglected. Though, methods have been developed and implemented to perform device authentication, such as physically unclonable features (PUFs), recent work has shown that the deterministic responses of PUFs can be replicated [15]. Therefore, a method by which to authenticate devices through non-deterministic and truly unclonable means is imperative. Additionally, because re-deploying networks of physical IoT devices with improved security systems is likely cost prohibitive in response to latent security vulnerabilities, more robust software-based device authentication techniques, such as radio frequency (RF) fingerprinting, which could be incorporated into the IoT device's layered defenses during production and could be updated remotely, are needed.

In the IoT-realm, low-cost, mass produced devices are often considerably resource-constrained. Consequently, our research aims to pave the way for a low-cost, low-power, and low-latency multi-factor authentication method utilizing neural networks (NNs). More specifically, this work investigates the feasibility of using NNs which cue on the minor, but unique, perturbations of physical (PHY) layer characteristics on the transmitted signals (RF fingerprints), for multi-factor authentication. When used in addition to other lightweight security schemes, such a solution would provide a robust,

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layered security approach appropriate for IoT devices [16].

To show feasibility, the use of NN-based SEI algorithms for IoT node authentication is examined using a NN-based SEI algorithm previously developed [17], [18]. NN-based authentication usage protocols for various device and network types are also discussed, and the approach given in [18] is executed on both a resource-rich and more resource-constrained device. The additional memory and computational costs associated with the granularity of such NN-based SEI methods will be examined in the context of the considered approach.

The results shown herein indicate that NN-based SEI algorithms will need further refining before implementation in today's large-scale systems, highlighting the challenges in dealing with NNs in IoT [19]. However, there are numerous benefits to integrating NN-based authentication techniques into the IoT. An approach like that in [18] is well-suited to IoT networks because of the large variation amongst low-cost, mass-produced nodes, caused by relaxed manufacturing tolerances. Such variances result in a highly diverse set of RF fingerprints which are difficult to impersonate based on their analog nature [20]. Using NN-based techniques to help identify and authenticate IoT using their RF fingerprints is particularly promising as such an approach reduces the need for human oversight. More specifically, a NN approach to device authentication can operate on the incoming raw IQ data, eliminating the use of *a priori*-determined, expert-defined features often used. This makes the approach more flexible and robust because feature extraction techniques are often only accurate over a limited parameter range, can be heavily effected by channel conditions, and can be inconsistent. In NN-based approaches, these effects can be far less severe, as they can be trained for. Operating on the raw IQ data stream can also lead to lower latency, as feature extraction can be time consuming, [21], [18] and computation would be reduced for transmitting devices because RF fingerprints are implicit in each transmission. As a result, NN-based SEI shows great promise in the rapidly growing field of IoT security and adds a much needed layer of security to these devices.

The rest of this paper is organized as follows: In Section II, recent works in the area of NN and other machine learning-based SEI algorithms are described. Sections III and IV detail current IoT security techniques and the IoT model used in this work. In Section V, current SEI techniques are discussed. Additionally, the NN-based SEI approach considered in this work is also presented and described in detail in Section V, including the NN design and algorithm performance.. NN-based SEI is discussed in the context of IoT in Section VI, and possible usage protocols are given. Section VII presents and discusses the results of the considered NN-based SEI algorithm's performance on different classes of devices for use in the IoT realm. Finally, directions for future work and conclusions are provided in Sections VIII and IX.

II. RECENT WORK

While machine learning (ML) and NN techniques provide the framework to eliminate the need for expert-defined features, recent works in ML and NN-based SEI have continued

to depend on these hand-crafted features. Most pertinent to this work, in [22], RF fingerprinting was used with deep learning NNs in cognitive radio networks to identify emitters using *error signals* (the difference between the received signal, which contains emitter specific effects, and the ideal transmitted signal), work by Patel *et al.* [23] created an RF-DNA fingerprint using statistical features (such as the variance and kurtosis, of the instantaneous amplitude, frequency, and phase of the received signal) which were matched to known emitters using ensemble classifiers, and recent work by Chatterjee *et al.* detailed RF-PUFs and their characteristics and used symbol-based NNs for node authentication [15]. However, in both [22] and [23], knowledge of the transmitted signal's modulation scheme or the transmitted signal itself is assumed, and the approaches are unable to identify emitters unseen during training [23]. Additionally, the work in [15] highlighted the impacts of outside factors such as temperature on the approach, indicating RF-PUFs are susceptible to environmental changes. Other work of interest, include [24], [25] and [26]. In these works, pre-defined expert features were used as input to NN or other ML-based SEI algorithms, such as infinite Gaussian mixture models (IGMMs), to calculate device proximity, perform session key establishment between nodes, distinguish and authenticate physical devices, and detect anomalies at a gateway-level.

III. SECURITY BACKGROUND

Many security practices focus primarily on information security areas of concern: *confidentiality*, *integrity*, and *availability* [27]. However, the IoT networks introduces new issues with additional complexities to security, due to their size and flexibility, as well as the use of low-cost devices. Therefore, device authentication is becoming an increasingly important element of the multi-factor security system, in addition to the current information security practices. Figure 1 highlights the Transmission Control Protocol and the Internet Protocol (TCP/IP) network stack and where the following security and communications protocols mainly reside. While current network security protocols such as Datagram Transport Layer Security (DTLS) [28] and Internet Protocol Security (IPsec)

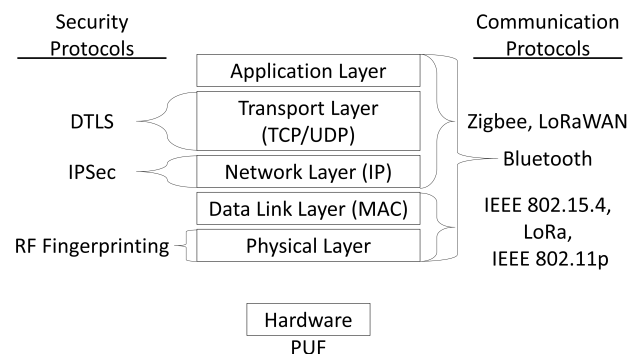


Fig. 1. A diagram of the TCP/IP network stack highlighting where different current security and communication protocols are mainly performed. Note that PUFs do not fall into the stack, so they are an ad hoc type of security feature.

[29] provide data security, both have been designed with the traditional Internet in mind, so the adaptation to IoT applications is not straightforward. Additionally, neither DTLS nor IPsec are lightweight enough to be executed on IoT devices, and asymmetric encryption's large key sizes, numerous handshakes, and heavy-duty encryption can slow down ideally quick, latency-critical links between nodes [30]. Therefore, more lightweight network protocols and security solutions are becoming prevalent in the literature and implemented in IoT-based networks and devices.

A. IoT Security

The communication and security schemes commonly used in IoT systems, described below, are often based on traditional security practices. While many of these security schemes provide data security, they do not provide a reliable method for performing device authentication. Additionally, many of these techniques and security schemes require multiple transmissions between devices and additional computational efforts from both devices to calculate the response to the challenge.

1) *Bluetooth*: Bluetooth is a short-range communication protocol that aims to provide low-power links amongst connected devices. Bluetooth relies on an initial pairing of devices with a shared secret value, which may be observed by an adversary, to allow a device to initially join a network. Bluetooth security is based on a combination of device ID, random number generator (RNG) value, personal identification number (PIN), and shared link key to perform authentication and other security features [31]. However, Bluetooth only verifies device authentication based on these values. Therefore, given the ability to watch the pairing process, attackers are able to retrieve and replicate the necessary values, achieving the synchronization necessary to spoof a device. Because the actual user of the device is never authenticated, the network is unable to distinguish one device from another, and the possibility of an attacker gaining access to the network remains. Successful spoofing and man-in-the-middle attacks on Bluetooth devices have been achieved in the past due to these vulnerabilities [32].

2) *Zigbee*: Zigbee [33] is a communication protocol standard that is based on the IEEE 802.15.4 wireless personal area network (WPAN) standard [34]. These WPANs allow for the manufacturing and operation of low-cost devices in wireless networks, making this protocol ideal for many IoT systems. Zigbee allows network administrators to broadcast unencrypted network keys to all devices when adding new devices to the network, creating a vulnerability for an attacker to take advantage and attempt to perform device impersonation or MITM attacks. Similar to Bluetooth, the Zigbee protocol performs data encryption and message integrity checks, but the protocol has no ability to perform actual unique device authentication. Moreover, practical attacks against Zigbee have been performed [35], [36], and a Python-based exploitation tool, KillerBee, has been open-sourced [37]. Zigbee also is based on a time-division multiple access (TDMA) carrier sensing multiple access with collision avoidance (CSMA/CA) protocol, which can be easily disrupted with timing attacks due

to the hidden node problem [38], to inexpensively implement DDoS attacks.

3) *Long Range Wide Area Network (LoRaWAN)*: LoRaWAN is a proprietary communication protocol designed for low-cost, low-power devices [39]. LoRaWAN is similar to the previous schemes in attempts to provide security to IoT networks. LoRaWAN utilizes 128-bit cryptographic keys for network formation and data encryption. It provides two methods for devices to join a network: Over-the-Air Activation (OTAA) and Activation by Personalization (ABP) [39]. In [40], a vulnerability in the ABP procedure was exploited through device physical tampering and allowed the security keys to be compromised. Then, the attackers were able to fully impersonate the compromised device and communicate on the network undetected due to the inability of the network to perform actual physical device authentication.

4) *IEEE 802.11p*: The IEEE 802.11p standard [41] is the basis for inter-vehicular communication and between cars and infrastructure at the PHY and MAC layers based on the normal WiFi standard. The purpose of this standard is to provide very fast communication capability due to the nature of vehicular movement. The downside of this fast communication is the lack of overall authentication, which must be handled at higher levels. As more smart vehicles and infrastructure become widespread, the use of this standard will grow, requiring fast authentication of devices.

B. Multi-factor Authentication

Because many of the previously described protocols focus on network formation and data security, true device identity is still a major concern. Therefore, multi-factor authentication (MFA) techniques must be utilized to increase trust and security in IoT networks [14], [42]. MFA can generally be achieved by performing at least two of the following methods: *what you possess* (e.g. a physical token or ID card), *what you know* (e.g. a password, PIN, or cryptographic key), or *who you are* (e.g. unique physical characteristics) [42]. In many cases, MFA is layered with the data encrypted through symmetric cryptography [27], which requires both the transmitter and receiver to know the shared encryption key. However, for remote, physically inaccessible nodes, it may not be possible to ensure the device's token will not be tampered with, making *physical possession* the less preferred MFA method. For example, some software packages require a physical key dongle plugged into a computer to authorize use, but this method would not be suitable for a remote device. Therefore, it is assumed that two-factor authentication for resource-constrained IoT nodes will be performed with a combination of *knowledge* (asymmetric or symmetric cryptography) and *physical characteristics*. The following sections highlight some of the current techniques used to perform authentication based on these physical properties.

1) *RF Fingerprinting*: Similar to the use of biometrics, such as retina scanners and fingerprints, devices impart their own unique analog modifications to signals, both on the transmitter and receiver ends. Due to manufacturing process variances, even the *same* devices can have minor differences

in their local oscillators and I-Q amplitude and phases imbalances. Therefore, identification of devices can be performed by using various signal analysis techniques to observe these differences [15], [16], [21], [22], [43].

One caveat to this approach is the degradation of RF features due to environmental factors such as temperature, pressure, water density, other atmospheric conditions, and component aging [44]. These factors may bound the actual performance of some RF fingerprinting methods. Therefore, special considerations should be made in the design of a network, in order to achieve desired performance. For instance, the number of nodes in the network may be capped due to the additional stored parameters for modeling different environmental cases.

2) *Physically unclonable features (PUFs)*: PUFs are ideally unique physical features embedded in the silicon components of the chip during the manufacturing process [45]. A PUF is typically a circuit that is designed to give a repeatable response to a stimulus for use in a challenge-response scenario. The actual manufacturing of this circuit introduces or exploits minor imperfections in the manufacturing process producing a unique signature to the actual response used for authentication, much like an intentional RF fingerprint. While PUFs are not a secure communication protocol, they have been implemented to provide physical device authentication. More specifically, a challenge/response scheme is used to determine proper authentication. In this practice, a device receives a challenge command that invokes a response from the PUF based on that specific command. Ideally, the correct response can only be generated by the PUF and not replicated by any other process. As previously discussed in other security applications, if the contents of a device's security scheme are compromised, an attacker is able to impersonate a device. PUFs attempt to solve this problem by providing an additional layer of authentication for devices.

However, though PUFs attempt to be unclonable, research has highlighted vulnerabilities with this technology. For example, it has been shown that some ring oscillator-based PUFs do not exhibit proper randomness for cryptographic functions [46], and ML-based attacks have been successfully used against 65 nm CMOS Arbiter PUFs [47]. Despite their shortcomings, researchers at MIT have done work with PUFs for authentication, and have worked to protect them against ML capabilities to learn the challenge/response pairs (CRPs) [48].

IV. IoT MODEL

In order to determine the best use of a NN-based SEI algorithm in IoT networks, assumptions for the network must be defined. The major components of the network are its configuration (size and topology) and its device makeup. These main factors will help bound the parameters needed for proper NN-based SEI in IoT.

A. Network Configuration

Most IoT networks are heterogeneous in nature, due to the different requirements of the systems involved. Therefore, for certain network configurations, a proper device hierarchy

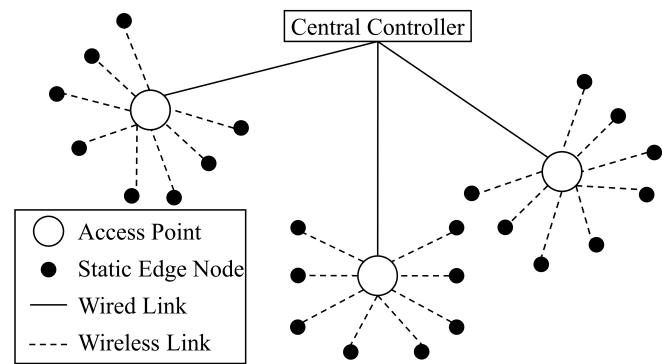


Fig. 2. Hierarchical star-of-stars network topology.

is implemented. This hierarchical design allows for easier control of network traffic because nodes are only able to communicate with other authorized devices. Also, networks with *a priori* knowledge of all devices allows an initial level of trust amongst nodes, given the ability to whitelist allowed connections. This hierarchical design can be used in many different topologies, such as star-of-stars and both sparsely- and densely-connected mesh networks, all of which will be further discussed in the following sections.

1) *Star-of-Stars*: The star-of-stars (also known as hub-and-spoke) network model, shown in Figure 2, is a very simple design. It allows for many edge nodes (ENs) to be connected to a single access point (AP), which is then directly connected to the central controller (CC). The routing is straightforward which eases initial implementation, but also causes inflexibility due to the minimum number of routes. Further, this topology can suffer when APs fail, unless sufficient resiliency considerations are made, such as configuring a backup AP for all nodes. However, this model is still ideal for very resource constrained ENs that have limited storage capabilities for routing tables as only the main AP and backup AP(s) would need to be stored.

2) *Sparsely-Connected Mesh*: The sparsely-connected mesh network is similar to the star-of-stars model in design, but allows for additional routing through specified relay devices. Although this adds more complexity to the routing, there are still hierarchical elements that help with determining known routes, and this added flexibility can help in networks prone to interference. Figure 3 shows an example of a sparsely-connected mesh network.

3) *Densely-Connected Mesh*: Figure 4 depicts an example of a densely-connected mesh network. Densely-connected mesh networks build on the concept of the sparse network, but allow for links between all devices within communication range. This greatly increases the complexity of storing routing information, but also provides the greatest amount of flexibility and resilience. As an added benefit, this type of network structure supports a self-healing capability. More specifically, when a link fails, a new route can be quickly reconfigured, minimizing downtime. Network monitoring also provides the ability to achieve low latency, by periodically calculating the fastest and most reliable routes and via wireless adaptations of time-sensitive networking [49]. However, due to the increase in

route management information, this topology may not be well-suited for very resource-constrained nodes depending on the size of the overall network. Densely-connected mesh networks can be compared to mobile ad hoc networks (MANETs) when the nodes are mobile. However, MANETs have even more stringent requirements due to their mobile-based structure and are out of scope for this paper.

B. Devices

1) *Edge Device*: The edge device is typically the most variable type of device in an IoT network. They are the major contributor to the heterogeneous nature of these networks. They can range from extremely resource-constrained, battery-powered or possibly energy-harvested sensors, actuators, or controllers to more capable devices with constant power supplies. As an example of a resource-constrained device, the MSP430FR5994 processor has 256 KB non-volatile memory, 8 KB RAM, and a maximum clock rate of 16 MHz [50], which severely limits an SEI implementation on this type of edge node. Therefore, the need to understand the lower bounds of these devices for SEI algorithms must be well understood.

Based on the possible network topologies described above, it may be assumed that a resource-constrained edge node would only need a very limited SEI implementation. For example, in a star-of-stars network, due to the well-defined and small number of links amongst devices, an edge node may only need to perform simple binary authentication; i.e., authentication decisions as to whether or not an incoming transmission belongs to its designated AP. However, as the networks grow more complex, the feasibility of SEI on the outer edge nodes becomes questionable. Because edge nodes only need to identify nearby neighbors that are common links (APs), shrinking the required number of stored parameters should help increase the feasibility of some SEI techniques on less capable devices.

2) *Access Point*: An access point in an IoT network will most likely be more capable than its edge nodes, and is typically more resource-rich with clock rates ≥ 1 GHz, RAM capacities of ≥ 256 MB, memory storage ≥ 1 GB, as well as a constant power supply. One example of this type of device is the Cisco wireless gateway for LoRaWAN, which boasts a 1.33 GHz processor, 1 GB of RAM, and 4 GB of flash memory

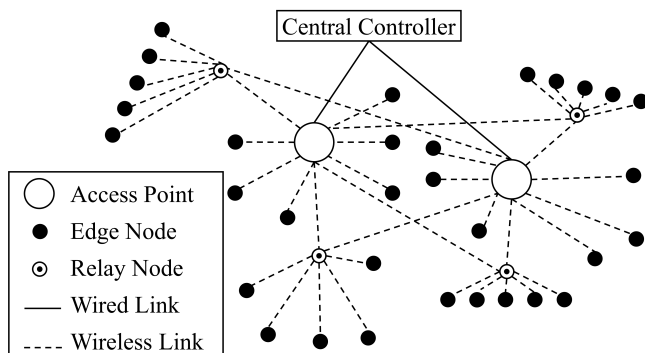


Fig. 3. Sparse mesh network topology.

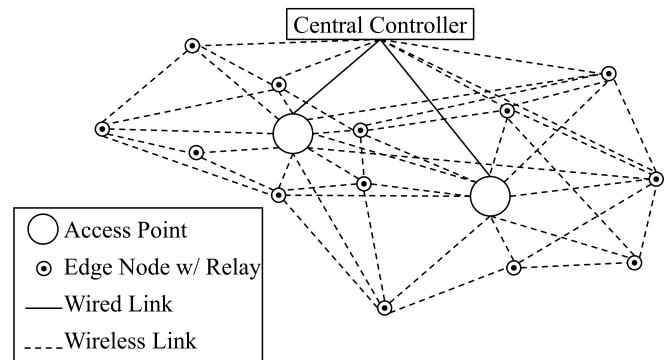


Fig. 4. Dense mesh network topology.

[51]. AP devices may be required to identify upwards of 100 devices on their subnet or combination of subnets (if used as a back-up). However, as more IoT gateways continue to hit the market with specifications similar to the Cisco wireless gateway, the realization of NN-based SEI implementation on these devices becomes very feasible, without significant network degradation or latency.

3) *Central Controller*: The central controller (CC) is usually the master device of the entire network. The CC may also act as a repository for all of the device's SEI parameters (properly safeguarded) that can then be securely transferred to new devices to achieve a type of transfer learning for the network. The CC can be a traditional desktop computer, a server farm (cloud), or may be composed of application-specific graphics processing units (GPUs), depending on application. Therefore, power, processor, and memory constraints of CCs will not need to impact the SEI design.

4) *Other*: As the IoT continues to grow, new device types may play bigger roles. For example, relay-type or other hybrid devices may be needed to improve network performance without the use of an additional gateway in order to lower costs. These devices may be more capable than current edge nodes, but will likely not be as resource-rich as APs. Additionally, these devices may have long dwell times before waking up to performing an expensive operation, which will impact how SEI may be performed through these devices. Due to the unknown nature of these devices, NN-based SEI methods will need to be flexible to allow for implementation on future node/device types.

V. SEI BACKGROUND

A. Traditional SEI Techniques

SEI, the act of matching a received signal to an emitter, has been performed in various domains, including the IoT [52], [15]. Early SEI techniques were used to determine the make and model of automobile engines using audio emissions and harmonics [53]. RF-based SEI techniques continue to be used in both commercial [54] and military [55] contexts to help in determining friend from foe. SEI or RF fingerprinting is possible because of the unique and unintentional characteristics that an emitter imparts onto each transmission, commonly caused by hardware imperfections and manufacturing inconsistency,

known as its RF fingerprint. Traditional SEI techniques utilize expert-defined features to capture these RF fingerprints in tandem with clustering algorithms used for classification and verification [55].

Expert-defined features are typically either extracted from the transient or steady state portion of the received signal. Features extracted from the transient portion of the received signal often examine the time-domain, frequency-domain, or phase-space [56]. For example, a popular frequency domain feature is the power spectral density [57]. The types of features extracted from the steady state signal vary widely including cyclic features [58], preamble periodicity [59], and wavelet based features [60].

The primary limitation of traditional SEI techniques is the use of these expert-defined features to characterize emitters of interest. Expert-defined features are often only accurate over a limited parameter range, can be heavily effected by channel conditions, and are reliant on accurate and consistent feature measurement [55], [61]. Further, expert-defined features only examine certain aspects of the received signal, and therefore the selection of features that characterize the emitters of interest is critical and can be time consuming. The extraction of features itself can also be extremely computationally expensive. In particular, transient processing typically requires high-speed over-sampling. Raw IQ based NN feature extraction techniques offer the potential for fingerprinting when critically sampled, lowering the computational expense.

B. NN Approaches to SEI

More recent work in SEI makes use of ML and NN-based techniques, but often continues to use expert-defined features as input to these algorithms, such as ensemble classifiers and neural networks, at the classification step [22], [23], [62]. While these approaches often yield good results, the continued reliance on expert-defined features means that the limitations outlined above are still highly relevant. Furthermore, these approaches are often only able to classify a known set of transmitters, failing to identify unknown emitters, and in the worst cases, labeling unknown transmitters as one of the learned classes.

The subset of NN techniques considered in this paper are convolutional NN (CNN)-based, and operate only using the received data, in raw IQ format, as input to the algorithm, relying on the CNN to objectively form a set of machine-learned features that enable accurate decision of the emitter. Two such approaches were developed in prior work [21], [18]. In the first approach, a CNN was trained to estimate the IQ imbalance of the transmitter [17]. By comparing the estimated gain offset to Gaussian distributions fitted to histograms of the CNN output, the estimated gain offset value could be used to identify emitters. Additionally, numerous estimates can be aggregated to produce more accurate results [18]. The second approach used a method called *supervised bootstrapping* to allow the CNN to learn emitter specific features from a known set of emitters. These features were fed to the density-based spatial clustering of applications with noise (DBSCAN) clustering algorithm for classification/identification. It was then shown

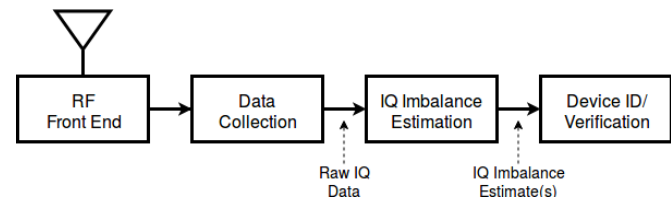


Fig. 5. The considered CNN-based SEI approach implemented on IoT devices.

that these features could be used to describe unknown emitters, allowing for the identification of these emitters using the clustering-based classification approach [21]. Both approaches also consider and handle the case of unknown emitters.

C. Considered Approach

In the following sections, we examine the use of NN-based SEI algorithms for IoT node authentication using the algorithm presented in [18] and shown in Figure 5 as proof of concept. While the protocols described below generalize to any NN-based SEI algorithm that utilize only raw IQ data as input, the approach considered here has several benefits lending itself to implementation on IoT devices. First, the approach is modulation agnostic, and it does not depend on a pre-trained classifier, and as a result, the approach has the potential to identify spoofers as well as the ability to whitelist new emitters in real-time. Second, because the approach uses the inherent feature learning abilities of CNNs rather than hand extracted features, the approach requires only 1024 raw IQ samples, or approximately 2 bytes worth of received data symbols [22], to identify a device. However, given more data, results can be aggregated to produce more accurate identification results. Once the IQ imbalance of the emitter has been estimated using the CNN, the identification and/or verification step is a simple and computationally inexpensive lookup. Finally, because IoT devices are often low-cost, it is likely they will exhibit more severe hardware impairments than high-end emitters, as previously discussed. Therefore, the approach will likely be able to uniquely identify more devices.

As briefly described above, the considered approach is CNN-based, and uses only the received signal, in raw IQ format, as input, which may be obtained using any blind signal detector, such as those found in [63]. Additionally, the algorithm operates without use of test signals, successive iterations, demodulation, or statistical measures, as is often required in traditional IQ imbalance estimation approaches. The considered approach is like the approach proposed in [15] in that both approaches identify emitters using IQ imbalance, a feature of the received signal caused by hardware non-idealities in the transmitter. More specifically, a CNN is used to estimate the gain offset of the transmitter, α , which is then compared to Gaussian distributions fitted to histograms of the CNN output to identify or verify emitters, the details of which can be found in [18].

While the considered approach and that given in [15] both identify emitters using their IQ imbalance, the considered approach uses only the raw IQ samples as input to the CNN,

TABLE I
CNN IQ IMBALANCE ESTIMATOR PARAMETERS

Layer	Size	Activation Fn.
Input	1024 IQ samples	-
Conv1D 1	12 filters of length 15	ReLu
Conv1D 2	19 filters of length 16	ReLu
MaxPooling	pool size = 2	-
FC 1	282 nodes	ReLu
FC 2	246 nodes	ReLu
FC 3	62 nodes	ReLu
Output	1 nodes	Linear

while the approach used in [15] is symbol-based, and therefore requires additional processing (i.e. synchronization) to obtain said symbols and requires significantly more data. Further, in the considered approach, the CNN is used to estimate the expert-defined feature, whereas in [15], expert-defined features are extracted prior to being fed to the neural network for identification. As a result, the approach in [15] is only able to identify prior white-listed nodes, and unknown and potentially harmful devices are falsely identified as known nodes.

1) *Neural Network Design*: The CNN used to estimate α , shown in Figure 6 and contains two 1D convolutional layers, a single max pooling layer, followed by four fully-connected layers. The final fully-connected 'layer' is just a single node which produces the output, a single value between $(-\infty, \infty)$, representing the estimated gain offset of the transmitter. The hyper-parameters of the CNN gain offset estimator are given in Table I. In total, the model contains 2.3 million trainable parameters. It is important to note that this architecture was developed and selected to achieve the high accuracy, without any attention paid to the size and complexity of the network. Therefore, the architecture could likely be simplified or altered in a variety of ways to reduce the number of trainable parameters and/or overall memory footprint of the trained model, with little to no performance loss, as discussed further in Section VIII.

2) *Performance*: The CNN was tested on datasets generated with the open-source *gr-signal_exciter* module in GNURadio [64], which were designed to best model gain and phase imbalances of real systems. The performance of the considered algorithm at various signal to noise ratios (SNRs) is shown in Figure 7. More specifically, Figure 7 shows the gain offset separation needed between emitters to achieve 80, 90, and 95% probabilities of correct classification. As expected, in lower SNR scenarios, the emitters of interest need to exhibit larger differences in gain offset, in order to maintain a high probability of correct classification. Additionally, as the probability of correct classification required by the system increases, so does the minimum gain offset separation needed between emitters. Further results in [18] also showed potential to improve the performance of this algorithm by narrowing the parameter space of the gain offset estimator network, as well as the ability to aggregate decisions across captures to improve performance. This indicates, for IoT implementation, that each sensor's configurable parameters (such as transmission power and physical placement) may need to be carefully designed and thoroughly tested to ensure that the proper gain offset

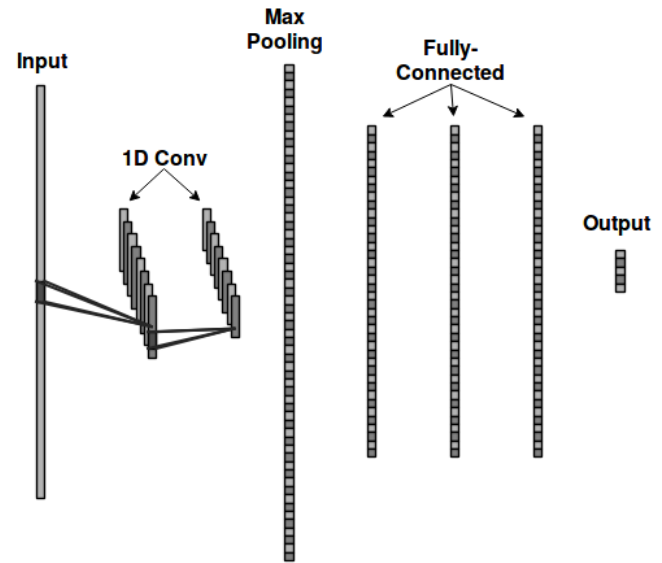


Fig. 6. The CNN model used for IQ imbalance estimation in the considered CNN-based SEI approach.

values are achieved.

3) *Memory Requirements*: In order to execute the proposed algorithm on an IoT device, the device must be capable of holding the weights and biases of the trained CNN in memory. The model used to produce the results shown in Figure 7 requires 18.7 MB of memory. This alone makes executing the proposed algorithm on the most resource constrained edge nodes currently infeasible. However, as previously mentioned, the selection of said model was purely performance-based, and no attention was made to the size of the model during training or testing.

VI. NN-BASED SEI FOR IOT

Although IoT nodes are often resource-constrained devices with very limited functionality, due to their low-cost nature and

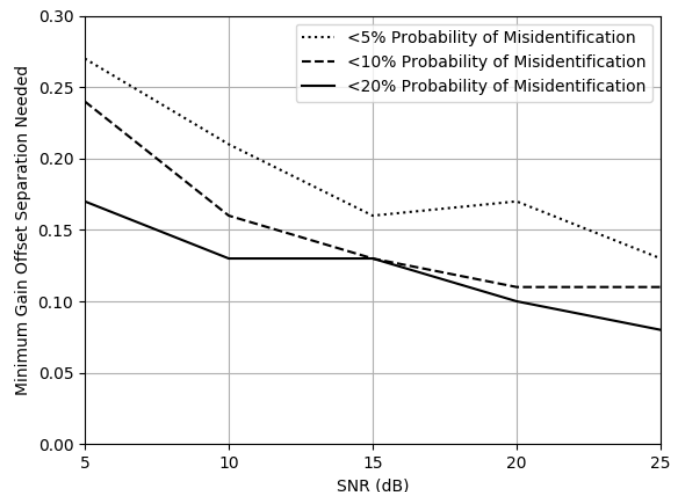


Fig. 7. Gain imbalance separation needed to achieve different levels of correct classification at different SNRs.

small footprint, these devices are ideal sensors and actuators that can be placed in remote, not easily accessible locations. Therefore, careful considerations must be made in the overall design of NNs for use on IoT devices. For example, IoT devices may be battery powered with minimal memory, limiting the number of NN trainable parameters that can be stored.

Generally, current NN or other ML applications in IoT are data-driven processes used to help improve overall confidence in and efficiency of existing systems, rather than as a component of the network stack. While there has been work in successfully implementing deep learning network processors on IoT-like devices [65], they are currently limited to tasks such as facial recognition due to restricted memory capacities (≤ 1 MB). As ongoing research continues to improve storage size [66] and overall IoT capabilities evolve, there will be more opportunities to implement NN-based algorithms on these devices.

The setup and usage of the a NN-based SEI algorithm on a network of IoT devices is dependent upon a variety of factors, including network configuration and device/system constraints. However, it may be assumed that all APs, relay nodes, and CCs will be capable of identifying N whitelisted devices and detecting unknown devices. Additionally, it should be noted that N may change dynamically over time as new devices join or leave the network.

As previously mentioned, edge nodes will typically be more resource-constrained than APs, relay devices, or CCs. In star-of-stars or sparsely-connected mesh networks, edge nodes will only need to verify a single AP or relay device. However, in a densely-connected mesh network, edge nodes may also need to be able to identify N devices as well.

As resource constraints are of primary concern when deploying a NN-based SEI algorithm on IoT devices, edge nodes in particular, it should be noted that the algorithm could be utilized in an IoT network in several ways. While running the SEI algorithm continuously would be most secure, it would also require the most power and memory and incur the most latency. Therefore, this is likely to only be feasible on a resource-rich device such as the AP or CC. A more likely use case is to run such an SEI algorithm periodically or on a triggered basis. This will be less secure, yet its frequency of initiation may be considered according to the value of the data being communicated. However, periodic or triggered SEI use is far more power efficient, and will incur far less latency in resource constrained systems. In the most resource-constrained systems (i.e. edge nodes), it may also be possible to rely on other devices for verification. This is clearly the least secure solution, as the verifying device will be assumed to be a trusted device at all times. However, this risk can be improved by strengthening the timing and distance profile of the verifying node. This solution will also incur the most latency, requiring the node to transmit to another device and await response. Given these trade-offs, three possible protocols for using a NN-based SEI algorithm in an IoT network are described below.

A. One-Way SEI

Given that it is currently improbable to perform SEI on a resource-constrained edge node, for certain networks, the

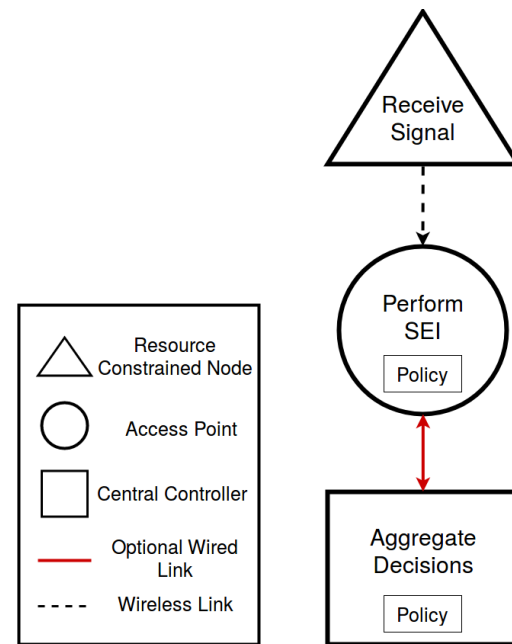


Fig. 8. One-way SEI approach.

option exists to only perform one-way authentication of an edge node to an AP. This method assumes accepted risk that an AP is unlikely to be compromised, reducing the need for mutual authentication to be performed by an edge node. This example is illustrated in Figure 8. After the AP performs SEI, it may have local policies in place to handle flags raised from unauthenticated devices, or it may relay that information to the central controller to perform further aggregation and decision policies.

B. Mutual Authentication Through a Secondary Device

Another approach, allowing mutual authentication for edge nodes, is to perform SEI on a secondary device that is physically located near the edge device, as shown in Figure 9. This secondary device would have more capabilities than the typical edge node, such as a Raspberry Pi Zero class device compared to an MSP430 class device, and would perform SEI on-demand (triggered or random) from a nearby edge node. This method allows the edge node to perform its normal operations while off-loading the computationally expensive authentication to a nearby, trusted device. In order to make it infeasible for an attacker to impersonate either of the devices, the distance and/or timing constraint between the edge node and secondary device will need to be well-defined. The benefit of this approach is that the secondary device needs no knowledge of the actual message because only raw IQ samples are needed as input to the proposed NN-based SEI algorithm. Therefore, the secondary device can simply perform the algorithm on the available transmissions from the AP. Additionally, depending on the network structure, a single secondary device could perform SEI for a small number of nearby edge nodes (≤ 5).

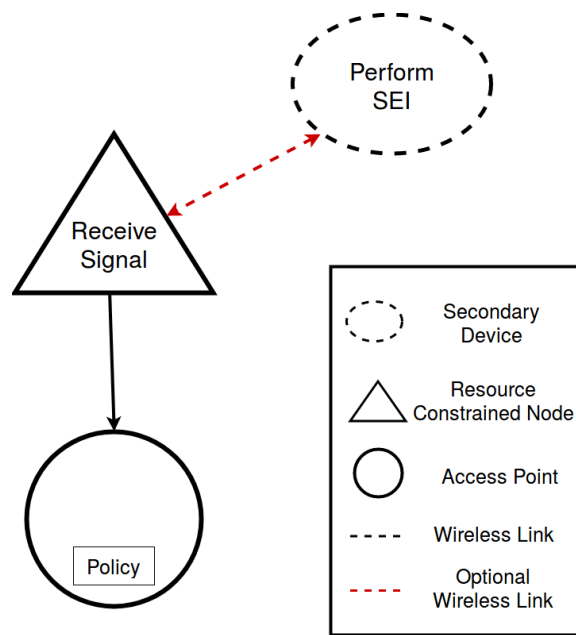


Fig. 9. Mutual authentication through secondary device approach.

C. Authentication-as-a-Service

A third approach, also permitting mutual authentication is the on-demand (triggered or random) wake-up of a moderately efficient, yet more capable, processor (e.g. ARM11 [67]) with non-volatile RAM by the primarily used microprocessor (e.g. MSP430). This method results in a relatively small duty cycle for the higher power processor, limiting its overall energy draw to the point of feasibility for an IoT edge node. This model is depicted in Figure 10.

VII. IOT IMPLEMENTATION RESULTS

Once the gain offset of an emitter has been estimated, the identification of said emitter is simply a lookup. Therefore the computational burden is in executing the CNN gain offset estimator. Therefore, this section describes the performance of gain offset estimation CNN portion of the considered approach, implemented in Python using the Keras and Tensorflow libraries [68], [69], on both a resource-rich quad-core Intel CPU [70] and a more resource constrained Raspberry Pi Zero [71]. The Raspberry Pi Zero is considered due to its low-cost and its ability to still run the CNN in Python. This device still may provide higher computing capabilities than battery-powered IoT devices, but it allows the ability to begin to bound the CNN performance in the IoT realm. Our goal is to move towards smaller systems on chips (SoCs) or more resource-constrained battery-powered devices as the CNN is further pruned to work in such devices.

The CNN used in the considered approach and described in Section V-C requires approximately 7.2 million multiplies and an equivalent number of adds to execute one signal capture containing 1024 samples. On a single thread of an Intel Xeon CPU E5-1620 v4 @3.5GHz processor [70], this requires 440 microseconds to execute. This highlights the feasibility of executing NN-based SEI algorithms on central controllers

or extremely resource-rich APs, with extremely low latency. However, the vast majority of edge nodes will not have this level of capability.

The same CNN model was also executed on a Raspberry Pi Zero [71], representing a more resource-constrained, edge node class of device, housing an ARM11 1 GHz processor with 512 MB of RAM and a removable SD card for non-volatile memory. On this class of device, the current evaluation was focused on the point-to-point link for an edge device's ability to perform SEI as a binary classifier (i.e., if the AP sent the message or not). Figure 11 shows a histogram of the execution runtimes of the model on the Pi. The mean execution time was approximately 123 ms with a mode of approximately 108 ms, indicating that this class of device is capable of performing NN-based SEI, with some latency incurred. However, it should be noted that authentication does not need to be performed on every burst, as was previously discussed in Section VI. Furthermore, the considered algorithm was not optimized for execution on such a device, and therefore could likely be altered to significantly reduce latency, as discussed below in Section VIII.

Given these results, it is clear that implementing the considered algorithm on the most resource-constrained devices, such as an MSP430FR5994, is not currently possible, nor is it likely to be possible in the near future. However, the algorithm, as currently implemented, could be integrated into an IoT network using one of the proposed protocols given in VI. Moreover, as IoT device capabilities improve, these results show the promise of using NN-based SEI algorithms, such as the considered approach, in networks containing approximately 100 nodes.

VIII. FUTURE WORK

Although results have shown the feasibility of using an NN-based SEI algorithm in IoT networks using the described implementation schemes, several areas have been highlighted as areas of future work to both improve such an algorithm and further test it's performance, increasing compatibility with IoT networks. These areas are further described below.

A. Implementation on Additional IoT Devices

The biggest factors for implementing a NN-based SEI algorithm are the memory footprint and clock rate of the device, therefore the designed NN must fit within allowed memory capacities and the algorithm must not add undue latency to the network. This work has shown that currently implemented NN-based SEI algorithms meets the specifications of many IoT access points and gateways. This is a critical junction in an IoT network due to the fact that these devices are more resource-rich and need to communicate amongst many other devices. As the NN-based SEI algorithm's footprint is reduced and edge node capabilities increase, there may soon be a common ground to implement a lightweight, but effective, algorithm onto some edge nodes to allow mutual authentication of devices without the need for expensive asymmetric cryptographic techniques.

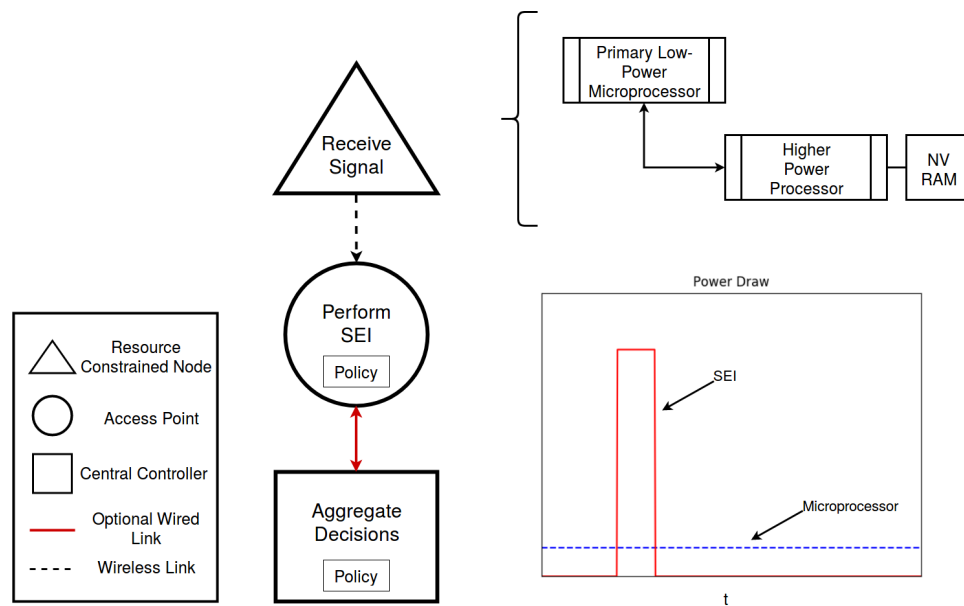


Fig. 10. The authentication-as-a-service model.

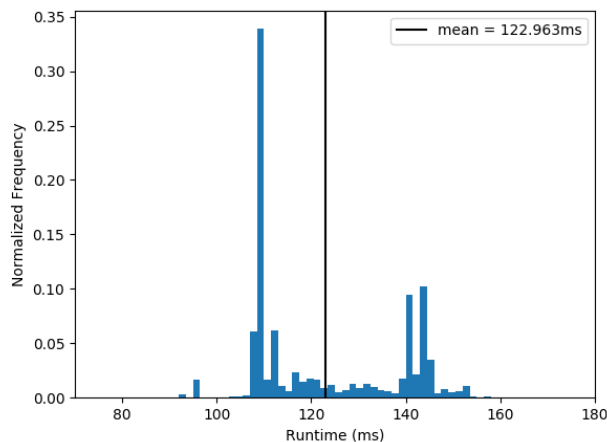


Fig. 11. Histogram of 10,000 SEI algorithm execution runtimes performed on Raspberry Pi Zero.

B. Decreasing Memory Requirements

It is clear that one of the primary limitations of using NN-based SEI approaches on IoT devices is memory constraints. For example, the CNN model developed to perform IQ imbalance estimation in [18] is far too large to be placed on an edge node, and may even be too large to be kept on some AP devices. Therefore, when developing NN-based SEI algorithms for use in IoT networks, attention needs to be paid to either significantly reduce the size of existing NN models or to the development of high performing, yet small NN models.

The easiest method to reduce the memory footprint existing NN models would be to retrain using 32-bit or even 16-bit floats, as opposed to 64-bit floats. Additionally, model pruning techniques such as those found in [72] and [73] may also be an effective method by which to reduce the size of NN models.

Training or retraining a NN while enforcing sparsity of the trainable parameters using methods such as those found in [74] and [75] is also a promising approach to reducing the size of existing or new NNs. Finally, ongoing research investigating the robustness of feed-forward classification NNs, such as multi-layer perceptrons (MLPs) and CNNs, to weight errors during training and testing has also shown that errors in the mantissa LSB have minimal performance impacts [76]. This result could be applied to further prune memory coefficient sizes.

C. Decreasing Latency

Another primary concern in utilizing NN-based SEI algorithms in IoT networks is the latency incurred as a result of running these algorithms. While latency can be controlled by executing the SEI algorithm less often, during critical communications only, the latency of currently implemented algorithms could be decreased in several ways. For example, the considered approach was implemented in Python, and would see significant speedups after porting into a more efficient language such as C. Additionally, the dependencies on open-source toolboxes, Keras and Tensorflow [68], [69], which may not be optimally implemented, could also be removed to further optimize the implemented algorithm. Efficient hardware implementations of neural networks is also an active area of research that could also be applied to both reduce latency and power consumption [77].

D. Further Testing

While the considered approach showed success using simulated RF data [18], the approach needs to be tested using real RF data to further validate the feasibility of using such an approach in an IoT setting. Future work looks to collect data from a custom PHY-layer [78] to highlight the agnostic

nature of the approach. The use of real data will provide better training sets and help refine the algorithm. Transfer learning between devices should also be explored, as this may help reduce overhead as new devices are added to networks. Further, the approach would need to be evaluated to determine its ability to scale to large IoT networks and its robustness to environmental impacts such as channel impairments, temperature, pressure, and hardware aging. Initial network scalability is currently being researched in order to help better understand these factors to better determine optimal network sizes.

E. SEI Algorithm Improvement

There are many ways in which the NN-based SEI can be improved, increasing the feasibility of implementing such algorithms on IoT devices. To increase the number of emitters the approach can identify, the neural network architecture could be further altered and optimized to achieve more accurate gain offset estimates, or the approach be extended to include the estimation of additional hardware impairments. However, it should be noted that improvements such as these may necessitate the use of larger or additional networks, requiring additional computation or memory at the IoT nodes.

IX. CONCLUSIONS

This paper has provided the basis for using a NN-based SEI algorithm as an additional layer of IoT device and network security. More specifically, the challenges of using NN-based SEI algorithms in IoT networks have been discussed, as well as design considerations and existing algorithm improvements to make using such algorithms feasible. This work has shown that an existing NN-based SEI algorithm can be integrated into systems as a device authentication technique, allowing multi-factor authentication in IoT without human-device interaction, with minimal latency on AP and CC class devices, as well as on edge node class devices, when used on an ad-hoc basis.

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